

# Statistics - Set 1

TMUA practice questions by topic. Specification M6.1-M6.4. 11 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The mean of five numbers is 12. When a sixth number is added the mean becomes 14. What is the sixth number?
- A** 24  
**B** 84  
**C** 20  
**D** 26  
**E** 16  
**F** 14
- 2 A histogram has a bar for the class  $10 \leq x < 15$  of height (frequency density) 3.2 and a bar for  $15 \leq x < 25$  of height 1.5. What is the total frequency of these two classes?
- A** 15  
**B** 23.5  
**C** 31  
**D** 4.7  
**E** 16  
**F** 47
- 3 The numbers 3, 7,  $x$ , 12, 18 (with  $7 \leq x \leq 12$ ) have mean equal to median. Find  $x$ .
- A** 12  
**B** 7  
**C** 10  
**D** 8  
**E** 11  
**F** 9

- 4 The table shows grouped data:  $0 \leq t < 10$ : frequency 4;  $10 \leq t < 20$ : frequency 10;  $20 \leq t < 40$ : frequency 6. Estimate the mean.
- A** 16  
**B** 18  
**C** 20  
**D** 15  
**E** 22  
**F** 17.5
- 5 Which of the following statements about a scatter graph showing strong positive correlation between hours of revision and exam mark is/are valid?
- I** Students who revised more tended to score higher.  
**II** Revising more causes a higher mark.  
**III** A line of best fit could be used to estimate the mark of a student who revised for a number of hours within the range of the data.
- A** none of them  
**B** I only  
**C** II only  
**D** III only  
**E** I and II only  
**F** I and III only  
**G** II and III only  
**H** I, II and III
- 6 A set of  $n$  numbers has mean  $m$ . Each number is doubled and then 3 is added. Which of the following is the new mean, and which the new range (if the original range was  $R$ )?
- A** mean  $2m + 3$ , range  $2R$   
**B** mean  $2m$ , range  $2R$   
**C** mean  $m + 3$ , range  $R$   
**D** mean  $2m + 3$ , range  $2R + 3$   
**E** mean  $2m + 3$ , range  $R$

- 7 A cumulative frequency graph for 80 observations passes through (20, 20), (30, 40) and (45, 60). Estimate the interquartile range.
- A 40
  - B 10
  - C 20
  - D 25
  - E 15
  - F 30
- 8 Class A has 20 students with mean mark 60; class B has 30 students with mean mark 70. What is the mean mark of all 50 students?
- A 68
  - B 65
  - C 62
  - D 67
  - E 64
  - F 66
- 9 For which of the following data sets is the mean greater than the median?
- A 1, 2, 3, 4, 5
  - B 5, 5, 5, 5, 5
  - C 2, 4, 6, 8, 10
  - D 1, 2, 3, 4, 20
  - E 1, 10, 11, 12, 13
- 10 The median of 7 distinct integers is 10 and the mean is 12. The smallest number is increased by 20. Which of the following must be true?
- A The mean and median both increase.
  - B The median increases and the mean does not change.
  - C The mean increases and the median does not change.
  - D The mean increases and the median may or may not change.
  - E Neither the mean nor the median changes.

- 11** A set of five distinct positive integers has median 10 and mean 10. Which of the following statements must be true?
- I** The largest value is at least 12.
  - II** The smallest value is at most 8.
  - III** The range is at least 4.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III